

## General Description

- Wireless temperature sensor designed for direct installation on busbars, cable terminations, contact points, etc., providing high measurement accuracy. Suitable for operation in harsh tropical and temperate climate conditions.
- Compact design, powered by self-powered energy harvesting or battery supply (battery life up to 10 years of continuous operation, with battery level indication). The sensor operates safely and reliably, with the ability to withstand lightning impulse and switching surge voltages up to 500kV.
- Wireless communication, with a transmission distance of up to 100 meters to the receiver unit in open environments.
- The sensor is suitable for both indoor and outdoor applications.



## Technical specifications

| Content                       | Technical Specification Details                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Communication                 | 2.4GHz bands                                                                                                                                                      |
| Transmission range            | 100 meters in open space                                                                                                                                          |
| Sampling Frequency            | 5s (self-powered); 60s (battery-powered)                                                                                                                          |
| Power Supply                  | Self-powered with primary current $\geq 1$ A or battery-powered (10 years)                                                                                        |
| Operating Voltage Rating      | Up to 500kV                                                                                                                                                       |
| Temperature Measurement Range | -10°C~120°C                                                                                                                                                       |
| Accuracy                      | $\pm 1^\circ\text{C}$                                                                                                                                             |
| Transmission Power            | 08 dBm                                                                                                                                                            |
| Working Temperature Capacity  | Temperature: $-20^\circ\text{C}$ to $80^\circ\text{C}$<br>Humidity: $\leq 99\%$ RH, non-condensing, suitable for near-sea environments<br>Altitude: up to 1,000 m |
| Partial Discharge Level       | $< 10\text{pC}$                                                                                                                                                   |
| Protection Level              | IP67                                                                                                                                                              |
| Environment Qualification     | RoHS 2                                                                                                                                                            |
| Size                          | Height: $\sim 14.4$ mm; Width: $\sim 31.5$ mm; Depth: $\sim 32.5$ mm                                                                                              |
| Weight                        | $\sim 35$ gram                                                                                                                                                    |
| Color                         | Black                                                                                                                                                             |

## Packaging

01 Sensor  
Included Accessories

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### Test Standards

1. Lightning impulse test in accordance with IEC 60060-1
2. Switching impulse test in accordance with IEC 60060-1
3. Water and dust ingress protection test in accordance with IEC 60529
4. Accuracy class or measurement uncertainty test in accordance with test method ĐLVN 138:2004
5. RoHS testing in accordance with IEC 62321-5, IEC 62321-7-2, IEC 62321-4, IEC 62321-6, IEC 62321-8
6. PD-free test up to a minimum voltage level of 500 kV in accordance with IEC 60270
7. Electromagnetic compatibility (EMC) testing: for the wireless sensor in accordance with QCVN 96:2015/BCT; for the data collection unit in accordance with QCVN 112:2017/BTTTT, QCVN 86:2019/BTTTT, and QCVN 54:2020/BTTTT
8. Thermal endurance test in accordance with IEC 60068-2-2:2007, with the sensor showing no deformation up to 160°C

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INVESTMENT CORPORATION JOIN STOCK COMPANY**  
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## General Description

- Temperature and humidity monitoring sensor with high accuracy, applicable in electromagnetic field environments up to 500 kV.
- Utilizes ultra-low-power technology, ensuring safe operation with a service life of up to 10 years.
- The sensor is intended for indoor use.



## Technical specifications

| Content                            | Technical Specification Details                       |
|------------------------------------|-------------------------------------------------------|
| Communication                      | 2.4GHz bands                                          |
| Transmission range                 | 100 meters in open space                              |
| Sampling Frequency                 | 140s                                                  |
| Environmental measurement range    | Temperature: -40°C to 125°C / Humidity: 0-100% RH     |
| Accuracy                           | Temperature: ±0.2°C / Humidity: ±2% RH                |
| Transmission power                 | 08dBm                                                 |
| Power supply                       | Battery-powered with a service life of up to 10 years |
| Operating voltage rating           | In environments with voltage levels of up to 500 kV   |
| Operating temperature and humidity | Temperature: -10°C to 80°C<br>Humidity: ≤ 100% RH     |
| Partial Discharge Level            | < 10pC                                                |
| Protection Level                   | IP 50                                                 |
| Environment Qualification          | RoHS 2                                                |
| Size                               | Height: ~13.3 mm; Width: ~34 mm; Depth: ~37 mm        |
| Weight                             | ~ 45 gram                                             |
| Color                              | Black                                                 |

### Packaging

01 Sensor  
Included Accessories

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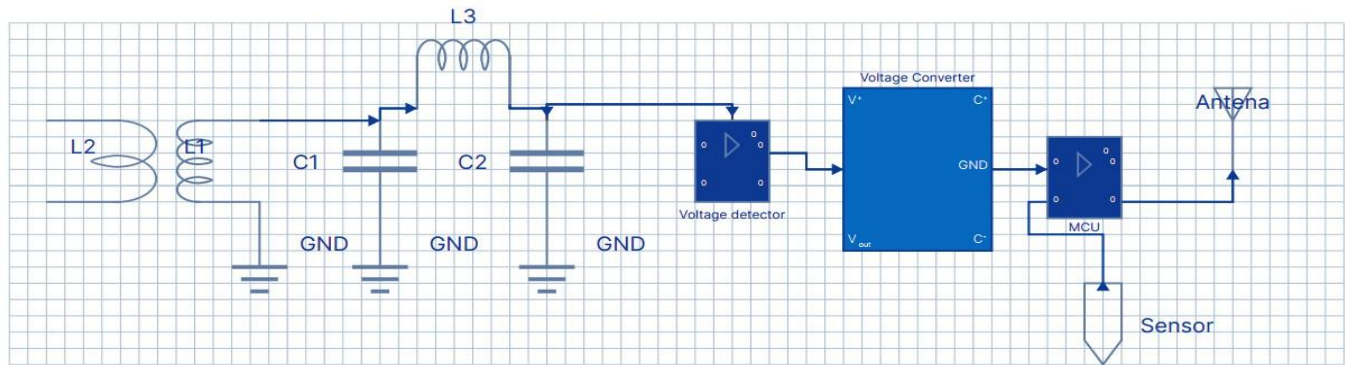
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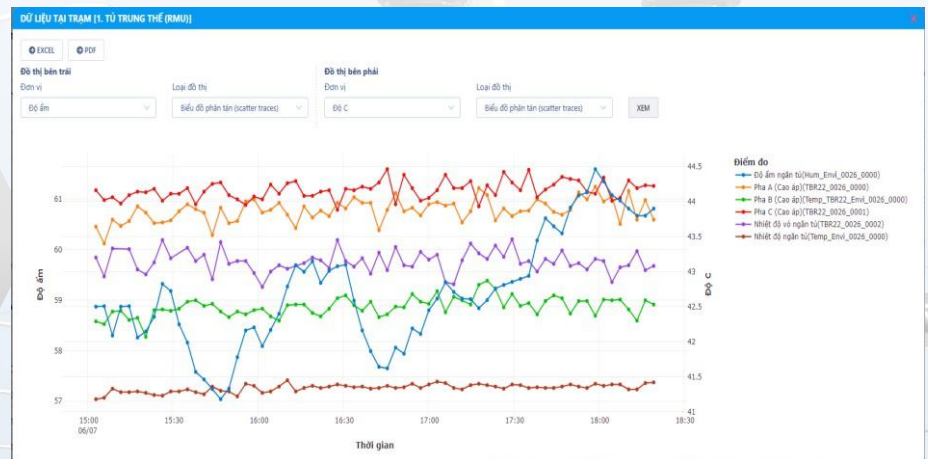
## Operating principle of the TBR22-BE sensor



## Central Monitoring Software

### Monitoring Software

Visualization, Analysis,  
Diagnosis, and Assessment of  
Electrical Equipment  
Condition



- Communication mechanism: Data is encrypted and transmitted via one-way communication. The sensor–receiver system is a closed and integrated system, operating without an operating system. It uses VTI’s proprietary protocol with dedicated hardware design, ensuring resistance to interference and cyber attacks.
- Real-time display of temperature values and alarm status of electrical equipment.
- Display of instantaneous temperature values at measurement points and monitored objects, with temperature trend charts (up to 10 years).
- Configurable alarm thresholds, providing abnormal temperature alarm lists for busbars, cable terminations, lugs, and contact points.
- User access management with multi-level authorization, capable of integration with third-party systems.
- Location positioning and retrieval of coordinates/locations on maps.
- Statistics and report generation with customization capability.
- Users can monitor the system via mobile applications on iOS and Android devices to promptly develop and implement response plans.

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## Overall Features

- Collects data from the wireless sensors. Utilizes 2.4GHz wireless technology with 16 channels allowing optimal data transmission speed.
- ISM band is used freely without requiring a license. Intergrates with a center or HMI according to specific standards or Modbus TCP/Modbus RTU standards, enhancing flexibility in industrial applications.
- Through a common RJ45 Ethernet form, physical connection will be sent to the center, making it easy to connect and configure using the user-friendly web GUI integrated into the device.



## Technical Specifications

| Specifications              | Detail                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Ready to integrate with existing monitoring systems at the station, such as Partial Discharge (PD) monitoring, Dissolved Gas Analysis (DGA), and Dielectric Loss monitoring (TD)...                                                                                                                                                                                                                |
| Function                    | Provides detailed identification of all monitored objects and their locations, optimizing the monitoring and operational processes.<br><br>Functions as an IEC 60870-5-104 master, supporting direct data reception from SCADA and displaying (numeric, alphanumeric) along with collected temperature, aiding in diagnostic and operational tasks.<br><br>Capable of connecting to SCADA systems. |
| Ethernet Interface          | * RJ45 - Used for configuration via the web GUI<br><br>Transmits data to the center.<br><br>Integrated with HMI.                                                                                                                                                                                                                                                                                   |
| Wifi Interface              | ** 2.4GHz 20dBm Used for configuration via the web GUI.<br><br>Transmits data to the center.<br><br>.                                                                                                                                                                                                                                                                                              |
| 4G Interface                | *** Micro simcard 4G LTE Network<br><br>Transmits data to the center.                                                                                                                                                                                                                                                                                                                              |
| RS485                       | <b>(Optional)</b> Modbus RTU<br><br>Transmits data to the center.<br><br>Integrated with HMI.                                                                                                                                                                                                                                                                                                      |
| Number of connected sensors | No hardware limitation. License limitation. Defaulted for 32 sensors.                                                                                                                                                                                                                                                                                                                              |
| Size                        | Height x Width x Thickness: 93 x 84 x 24 (mm)                                                                                                                                                                                                                                                                                                                                                      |
| Weight                      | 200 gram                                                                                                                                                                                                                                                                                                                                                                                           |

### Package

- 01 Gateway
- 01 Atena Wifi/4G (depend on Interface options)
- 1 AC/DC 1A Adapter

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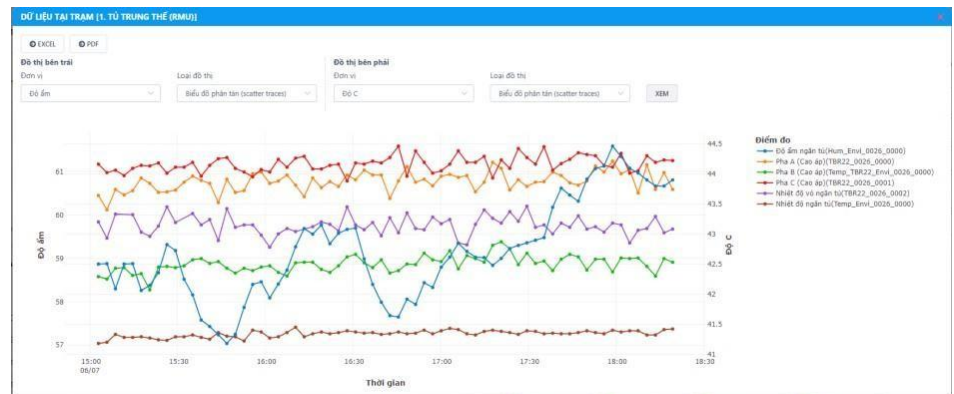
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## Centralized Monitoring Platform

### Monitoring Platform

Visualization, Analysis,  
Diagnosis, and Assessment  
of Electrical Equipment  
Condition



- Display real-time temperature values and warning statuses of electrical equipment.
- Show instantaneous temperature values at measurement points, and measured objects, and draw temperature trend charts.
- Provide a list of abnormal temperature warnings for bars, cable heads, contact points, etc., supporting maintenance work, periodic inspections without power interruption, saving time, increasing operator safety, and minimizing potential risks of heat-related fires.
- Locate and retrieve coordinates/positions on a map.
- Statistics and report generation with customization capabilities.
- Users can monitor through a mobile application (iOS/Android) for timely decision-making and action planning.

#### Information Security:

- Integrated sensor and gateway system without using the operating system, ensures no external attacks possible. The system will be destroyed in case of intentional hardware tampering for data retrieval.
- One-way (encrypted) communication from the sensor to gateway: Uses a proprietary protocol, preventing interception, attacks, data exploitation, or manipulation.
- Gateway sends one-way messages to the destination: Connects to a single server in the network using standard TCP/IP protocols such as MQTT, HTTP, HTTPS, FTP. Supports authentication using username and password or ".pem" encryption keys.

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